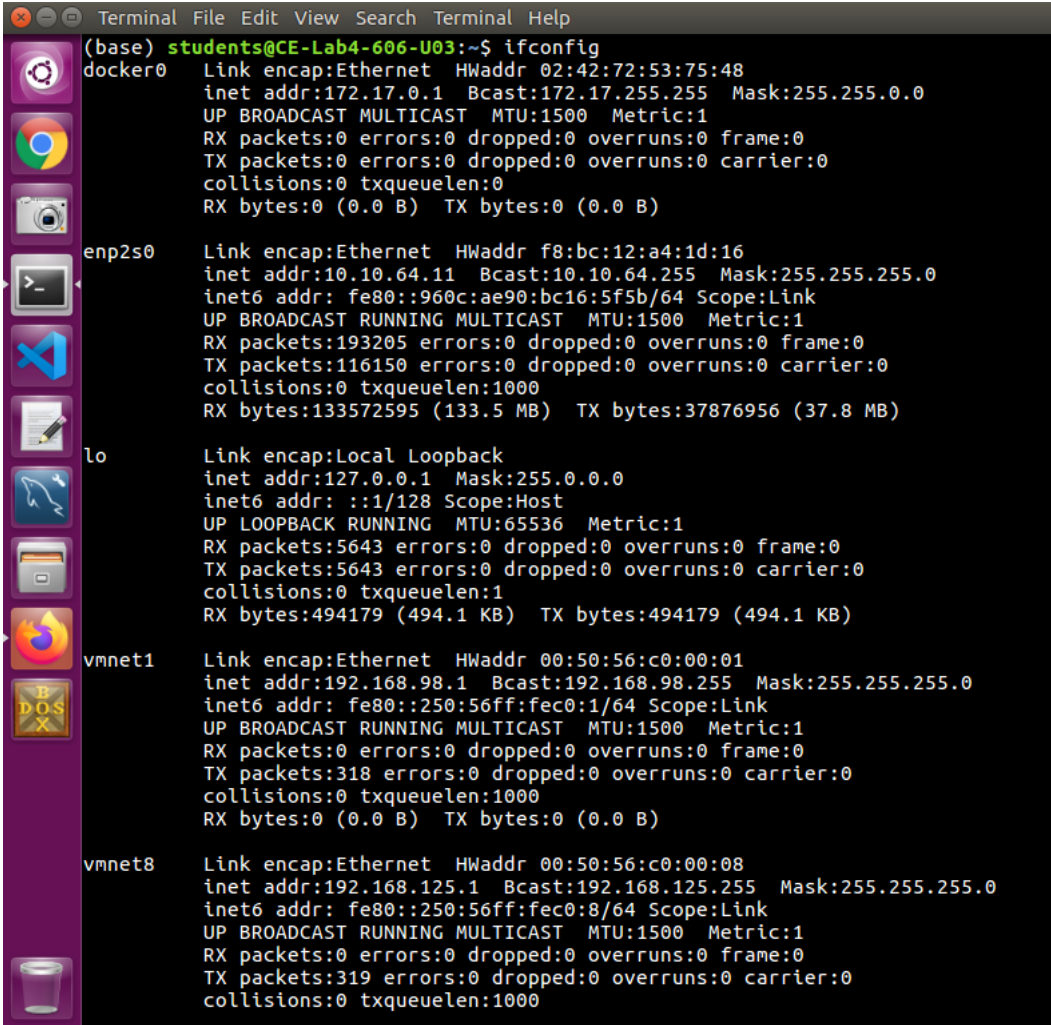


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TITLE:	Network Setup: Network configuration, commands and troubleshooting
PROGRAM:	Problem 1: To display and analyze the network configuration details of the system. Command used: ifconfig/ipconfig  The screenshot shows a terminal window with the command 'ifconfig' executed. The output displays details for several network interfaces: docker0, enp2s0, lo, vmnet1, and vmnet8. Each interface listing includes its link type (Ethernet or Local Loopback), hardware address (HWaddr), IP address (inet addr), broadcast address (Bcast), netmask (Mask), and various statistics like MTU, Metric, RX/TX packets, errors, dropped, overruns, carrier, collisions, and queue lengths. The 'lo' interface is a local loopback with address 127.0.0.1. The 'vmnet1' and 'vmnet8' interfaces are virtual network interfaces with IP addresses 192.168.98.1 and 192.168.125.1 respectively. Explanation:

Common terms --

- **Link encap:** Ethernet
→ Means the network uses Ethernet technology.
- **HWaddr**
→ Hardware address (MAC address) of the network card.
→ It is a unique physical address.
- **inet addr**
→ IP address (IPv4) of that interface.
→ Example: 10.10.64.11
- **Bcast**
→ Broadcast address used to send data to all devices in the network.
- **Mask**
→ Subnet mask.
→ It defines the network portion and host portion of the IP address.
- **UP**
→ Interface is active.
- **RUNNING**
→ Interface is ready to transmit and receive data.
- **MTU (Maximum Transmission Unit)**
→ Maximum size of data packet that can be sent at once.
- **RX packets**
→ Number of packets received.
- **TX packets**
→ Number of packets transmitted (sent).
- **RX bytes / TX bytes**
→ Amount of data received and sent.
- **enp2s0**
 - Main network interface
 - Has IP 10.10.64.11
 - This is your actual internet connection
- **lo (Loopback)**
 - IP 127.0.0.1
 - Used by the system to talk to itself
 - Always present
- **docker0**
 - Used by Docker
 - Virtual interface
- **vmnet1 / vmnet8**

- Used by VMware
- Virtual networks for virtual machines

Problem 2: To test the connectivity between the local system and a remote host.

Command used: ping google.com

```
(base) students@CE-Lab4-606-U03:~$ ping google.com
PING google.com (142.250.71.110) 56(84) bytes of data:
64 bytes from pnbomb-ad-in-f14.1e100.net (142.250.71.110): icmp_seq=1 ttl=117 time=2.32 ms
64 bytes from pnbomb-ad-in-f14.1e100.net (142.250.71.110): icmp_seq=2 ttl=117 time=2.28 ms
64 bytes from pnbomb-ad-in-f14.1e100.net (142.250.71.110): icmp_seq=3 ttl=117 time=2.23 ms
64 bytes from pnbomb-ad-in-f14.1e100.net (142.250.71.110): icmp_seq=4 ttl=117 time=3.12 ms
64 bytes from pnbomb-ad-in-f14.1e100.net (142.250.71.110): icmp_seq=5 ttl=117 time=3.05 ms
64 bytes from pnbomb-ad-in-f14.1e100.net (142.250.71.110): icmp_seq=6 ttl=117 time=2.36 ms
64 bytes from pnbomb-ad-in-f14.1e100.net (142.250.71.110): icmp_seq=7 ttl=117 time=2.23 ms
64 bytes from pnbomb-ad-in-f14.1e100.net (142.250.71.110): icmp_seq=8 ttl=117 time=2.32 ms
64 bytes from pnbomb-ad-in-f14.1e100.net (142.250.71.110): icmp_seq=9 ttl=117 time=2.37 ms
64 bytes from pnbomb-ad-in-f14.1e100.net (142.250.71.110): icmp_seq=10 ttl=117 time=2.33 ms
64 bytes from pnbomb-ad-in-f14.1e100.net (142.250.71.110): icmp_seq=11 ttl=117 time=2.29 ms
64 bytes from pnbomb-ad-in-f14.1e100.net (142.250.71.110): icmp_seq=12 ttl=117 time=2.25 ms
64 bytes from pnbomb-ad-in-f14.1e100.net (142.250.71.110): icmp_seq=13 ttl=117 time=2.35 ms
64 bytes from pnbomb-ad-in-f14.1e100.net (142.250.71.110): icmp_seq=14 ttl=117 time=2.26 ms
^C
--- google.com ping statistics ---
14 packets transmitted, 14 received, 0% packet loss, time 13019ms
rtt min/avg/max/mdev = 2.234/2.416/3.129/0.280 ms
(base) students@CE-Lab4-606-U03:~$
```

Explanation:

Ping checks connectivity and delay between the source and destination.

- 14 transmitted → Packets sent
- 14 received → Replies received
- 0% packet loss → Very good network connection
- time 13019ms → Total test duration

rtt min/avg/max/mdev = 2.234/2.416/3.129/0.280 ms

min → Minimum time taken

avg → Average time

max → Maximum time

mdev → Variation in delay

Low values indicate stable and fast connection.

Problem 3: To display active network connections and listening ports.

Command used: netstat

```
(base) students@CE-Lab4-606-U03:~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:1277          localhost:52010         ESTABLISHED
tcp        0      0 10.10.64.11:36666      relay-90babe9a.net:http ESTABLISHED
tcp        0      0 10.10.64.11:38598      lcbomp-in-f94.1e1:https TIME_WAIT
tcp        0      0 10.10.64.11:56558      123.208.120.34.bc:https ESTABLISHED
tcp        0      0 10.10.64.11:44564      del11s21-in-f10.1:https ESTABLISHED
tcp        0      0 10.10.64.11:60458      pnbomb-ab-in-f3.1:https ESTABLISHED
tcp        0      0 10.10.64.11:34430      hkg12s10-in-f4.1e:https ESTABLISHED
tcp        0      0 10.10.64.11:56284      tsa01s11-in-f3.1e1:http TIME_WAIT
tcp        0      0 10.10.64.11:41482      lcboma-as-in-f1.1:https TIME_WAIT
tcp        0      0 10.10.64.11:44566      del11s21-in-f10.1:https ESTABLISHED
tcp        0      0 10.10.64.11:42682      tsa01s11-in-f14.1:https TIME_WAIT
tcp        0      0 10.10.64.11:53586      sin11s28-in-f3.1e:https ESTABLISHED
tcp        0      0 10.10.64.11:41154      10.10.64.24:netbios-ssn ESTABLISHED
tcp        0      0 10.10.64.11:34422      hkg12s10-in-f4.1e:https ESTABLISHED
tcp        0      0 10.10.64.11:36590      del12s11-in-f14.1:https ESTABLISHED
tcp        0      0 10.10.64.11:56276      tsa01s11-in-f3.1e1:http TIME_WAIT
tcp        0      1 10.10.64.11:59216      172.16.10.20:http      SYN_SENT
tcp        0      0 localhost:52010        localhost:1277         ESTABLISHED
tcp        0      0 10.10.64.11:34900      93.243.107.34.bc.:https ESTABLISHED
tcp        0      0 10.10.64.11:59288      del11s22-in-f14.1:https ESTABLISHED
tcp        0      0 10.10.64.11:41706      lcboma-aw-in-f8.1:https ESTABLISHED
tcp        0      0 10.10.64.11:58488      bom07s26-in-f2.1e:https ESTABLISHED
tcp        0      0 10.10.64.11:56336      tsa03s08-in-f14.1:https TIME_WAIT
tcp        0      0 10.10.64.11:37804      a23-193-114-56.de:https ESTABLISHED
tcp        0      0 10.10.64.11:53636      lcboma-be-in-f14.:https TIME_WAIT
tcp        0      0 10.10.64.11:48480      pnbomb-bk-in-f10.:https ESTABLISHED
tcp        0      0 10.10.64.11:45324      151.101.193.91:https   ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags               Type           State         I-Node  Path
unix    2      [ ]                DGRAM                    33847    /run/user/1001/systemd/notify
unix    2      [ ]                DGRAM                    16098    /var/spool/postfix/dev/log
unix    2      [ ]                DGRAM                    11328    /run/systemd/cgroups-agent
unix    2      [ ]                DGRAM                    11333    /run/systemd/journal/syslog
unix   22      [ ]                DGRAM                    11342    /run/systemd/journal/dev-log
unix    9      [ ]                DGRAM                    11347    /run/systemd/journal/socket
unix    3      [ ]                DGRAM                    11327    /run/systemd/notify
unix    3      [ ]                STREAM            CONNECTED        186865
unix    3      [ ]                STREAM            CONNECTED        33770    /var/run/dbus/system_bus_socket
unix    3      [ ]                STREAM            CONNECTED        31647    @/tmp/dbus-VACZ0BJH2j
unix    3      [ ]                STREAM            CONNECTED        16101    /var/run/dbus/system_bus_socket
```

Explanation:

The netstat command provides information about current TCP/UDP connections, routing tables, and network interface statistics. It is useful for monitoring network activity and diagnosing network issues.

Internet Connections (TCP)

- Shows:
 - Local Address: 10.10.64.11
 - Foreign Addresses: Google, YouTube, HTTPS servers
 - State:
 - ESTABLISHED → active connection
 - TIME_WAIT → connection closed, waiting to clear

- SYN_SENT → connection attempt in progress

UNIX Domain Sockets

- Used for inter-process communication inside Linux.

Problem 4: To obtain domain name and DNS-related information.

Command used: nslookup google.com

```
(base) students@CE-Lab4-606-U03:~$ nslookup google.com
Server:          127.0.1.1
Address:         127.0.1.1#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.77.46

(base) students@CE-Lab4-606-U03:~$
```

Explanation:

Nslookup is used to query DNS servers to resolve domain names into IP addresses and vice versa. It helps in diagnosing DNS-related issues.

- DNS Server: 127.0.0.1
 - Indicates a local DNS resolver (systemd-resolved).
- Non-authoritative answer:
 - Domain: google.com
 - IP Address: 142.250.77.46

Meaning

- DNS successfully translated the domain name into an IP address.
- “Non-authoritative” means the answer came from a cached or recursive DNS server, not Google’s main DNS server.

Problem 5: To trace the path taken by packets from the local system to a destination host.

Command used: traceroute spit.ac.in

```
(base) students@CE-Lab4-606-U03:~$ traceroute spit.ac.in
traceroute to spit.ac.in (172.16.10.2), 30 hops max, 60 byte packets
 1  10.10.64.1 (10.10.64.1)  0.331 ms  0.323 ms  0.324 ms
 2  194.0.0.1 (194.0.0.1)  0.131 ms  0.142 ms  0.132 ms
 3  172.16.35.240 (172.16.35.240)  0.411 ms  0.412 ms  0.407 ms
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
```


	Explanation: Traceroute displays the path taken by packets from the local system to the destination ▪ Each numbered line represents a router (hop). ▪ The IP address and hostname of each router is shown. ▪ Three time values per hop (e.g., 0.462 ms 0.398 ms 0.361 ms) appear because: • Traceroute sends three probe packets per hop • Each packet may experience different delays due to: ○ Network congestion ○ Router processing time ○ Load balancing Why are multiple different times shown per router in traceroute output? Why are the times different even though the router is the same? <u>Answer:</u> Traceroute sends more than one packet (usually three) to the same router. Each packet measures how long it takes to go to the router and come back. That is why multiple time values are shown for one router. Even for the same router, the time can be different because: 1. Network traffic changes ○ At that moment, the router may be busy handling other data. 2. Processing delay ○ The router may take slightly more or less time to reply to each packet. 3. Packets may take slightly different paths ○ Some networks use load balancing, so packets may not follow the exact same path. Because of these small variations, each packet returns in a slightly different time.
CONCLUSION:	This experiment demonstrated the use of essential network diagnostic commands to analyze system configuration, test connectivity, resolve DNS queries, monitor active connections, and trace packet routes, thereby enhancing practical understanding of network troubleshooting.